

* Defn A set A is called a proper subset of set B if $A \subset B$ and $A \neq B$. We write $A \subset B$

* Defn $\mathbb{N}$: The set of natural numbers
$$\mathbb{N} := \{1, 2, 3, \dots\}$$

* Defn Set theoretic difference of A and B / The complement of B relative to A .
$$A \setminus B := \{x: x \in A \text{ and } x \notin B\}$$

* Defn Sets A and B are called disjoint if
$$A \cap B = \emptyset$$

* Lemma
$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$
$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$

* Theorem (Principle of Induction). Let $P(n)$ be a statement depending on the natural number n . Suppose that

① $P(1)$ is true

② If $P(n)$ is true then $P(n+1)$ is true

Then $P(n)$ is true for all $n \in \mathbb{N}$

Hint: Use well ordering property of $\mathbb{N}$

* Defn Well ordering property of $\mathbb{N}$. Every non empty set of $\mathbb{N}$ has a smallest element.

* Defn Let A and B be sets. The Cartesian product is the set of tuples defined by

$$A \times B := \{(x, y) : x \in A, y \in B\}$$

domain of f codomain of f

* Defn A function $f: \overset{\text{domain of } f}{A} \rightarrow \overset{\text{codomain of } f}{B}$ is the subset f of $A \times B$ st for each $x \in A$ $\exists$ a unique $y \in B$ for which $(x, y) \in f$. We write $f(x) = y$

$$\underset{\text{range of } f}{R(f)} := \{y \in B : \exists x \in A \text{ st } f(x) = y\}$$

* Defn $f: A \rightarrow B$. Image / Direct Image of $C \subset A$
 $f(C) := \{f(x) \in B : x \in C\}$
 Inverse Image of $D \subset B$
 $f^{-1}(D) := \{x \in A : f(x) \in D\}$

* Lemma $f: A \rightarrow B$. Let C, D be subsets of B then
 $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$
 $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$
 $f^{-1}(B \setminus C) = A \setminus f^{-1}(C)$

* Lemma $f: A \rightarrow B$. Let C, D be subsets of A . then
 $f(C \cup D) = f(C) \cup f(D)$
 $f(C \cap D) \subset f(C) \cap f(D)$

* Defn Given a set A , a binary relation on A is a subset $R \subset A \times A$ which consists of those pairs where the relation holds. Instead of $(a, b) \in R$ we write $a R b$.

* Defn Let A and B be sets. We say A and B have the same cardinality when there exists a bijection $f: A \rightarrow B$.

* Defn If A have same cardinality as $\{1, 2, 3, \dots, n\}$ for some $n \in \mathbb{N}$, we write $|A| := n$. If A is empty we write $|A| := 0$.
 equivalence class of all sets with same cardinality of A } finite cardinality

* Defn We write $|A| \leq |B|$ if there exists an injection from A to B .

* Defn If $|A| = \aleph$ we say A is countably infinite.

* Defn If A is finite or countably infinite we say A is countable.

* Defn The power set of A , denoted by $\mathcal{P}(A)$, is the set of all subsets of A .

* Theorem Let A be a set. Then $|A| < |\mathcal{P}(A)|$. In particular there exists no surjection from A onto $\mathcal{P}(A)$.